Edge Cases (Extended) - Voice Assistant on ESP32

Group A2, IoT course, Summer 2026.

The original list is edge-cases.md. This list extends it in the format of the course's robustness presentation, and adds the edge cases of the submission version (no SD card, Offline, switching networks, working without a computer).

For each case: what the user sees or hears (indication), and how the system keeps working (recovery). The last column says what the presenter does in the room to make it happen. The numbers in the demo plan refer to this list. All the demos run on the phone's hotspot.

Network and connection

#	The case	Indication to the user	Recovery	How to demo it	✓
1	The Wi-Fi drops in the middle of work	A red Wi-Fi light, the circle shows **Offline** (`no Wi-Fi`). With status sounds: two falling tones	The ESP32 reconnects on its own; if the network is still down a minute after the last attempt, it starts over. When the network comes back, the connection to OpenAI reopens within about 1-3 seconds (the OpenAI light is orange, `Wi-Fi is back - reconnecting` , until it is ready). If OpenAI still does not answer after 10 seconds, it goes back to retrying every 30 seconds. The timer and the controls keep working	On the phone, turn off the hotspot for 30 seconds and turn it back on. Wait up to about 15 seconds from when it came back: a green light, Ready, and two rising tones	
2	No Wi-Fi when the board is turned on	A red Wi-Fi light with the reason (`network not found` / `check the password`), Offline	The boot continues after 20 seconds. The connection to OpenAI opens on its own when the network comes up	Turn off the hotspot on the phone and press Reset (EN) on the board. Wait for the board to start without a network (up to 20 seconds), then turn on the hotspot and see that it connects on its own	
3	The network changed (a different hotspot, a different password)	The **Wi-Fi network** card shows which network the board is using	Enter the name and password and click Save & connect. It is saved in flash, without compiling. **Back to default** returns to the network from `SECRETS.h` . A saved network that fails twice in a row (for example a wrong password) is replaced on its own by the network from `SECRETS.h` (in memory only; `@@WIFI_NET` shows `default` , and the saved network stays in flash), and back the same way	Turn on a hotspot on a second phone (2.4GHz), enter its name and password in the Wi-Fi network card and click Save & connect. At the end, Back to default returns to the first phone	
4	There is Wi-Fi but OpenAI is not available (no internet, an invalid key)	A red OpenAI light `offline - retrying every 30 s` , the circle shows Offline (`OpenAI unreachable`). A key that OpenAI refuses: `API key refused (HTTP 401)` on the light and `OpenAI refused the API key` in the circle (case 35)	After 5 attempts (with a growing wait) it keeps retrying every 30 seconds, never stopping. A connection that was open when the internet disappeared is detected by the heartbeat (a ping every 15 seconds, 2 answers that did not come back within 10 seconds). Talk answers `offline`	On the phone, turn off mobile data and leave the hotspot on, so the board is connected to a network with no internet. If the phone turns off the hotspot together with the data, demo case 1 instead	
5	The connection to OpenAI drops in the middle of a question	`LISTEN #N FAILED: disconnected before a reply arrived` , the circle shows Reconnecting	It reconnects on its own, and the attempt counter resets after every successful connection. The disconnect fails only this question, and Hands-free continues	Ask a question, and as soon as it is sent turn off the hotspot before the answer starts, and turn it back on after a few seconds	

#	The case	Indication to the user	Recovery	How to demo it	✓
6	The server does not answer	`LISTEN #N FAILED: no server activity for 15000 ms`	After 15 seconds without any message the question fails, and the board closes the connection and reopens it (the conversation context, for example the name you said, is lost), then goes back to Ready. It does not get stuck	It cannot be caused on purpose (it is the server staying silent). Explain it verbally, and if there is a saved log where it happened, show it with Replay a log…	
7	The timer ends when there is no connection	The board beeps 3 times and blinks	The timer runs on the board itself. Only the spoken message needs a connection, and without one it is not said	Say `Set a timer for 30 seconds`, turn off the hotspot on the phone, and wait for the beep	

Hardware

#	The case	Indication to the user	Recovery	How to demo it	✓
8	No SD card at boot	A red SD light `no card (or not FAT32) - running without recordings`. The recordings tab: `no SD card`	The assistant works normally, without recordings	Remove the SD card from the board, press Reset (EN) and ask a question	
9	The control center disconnects (USB cable)	A red banner in the control center: `Lost the board's USB connection.`	The board is powered by the same cable, so when it is disconnected the board turns off, and it restarts when it is connected again (about 10 seconds until the opening greeting, and the settings are kept in flash). When the cable comes back, the page reconnects on its own and asks for the whole state (`STATUS`)	Unplug the USB cable from the computer, see the red banner, and plug it in again. The page reconnects on its own	
10	Board Reset	The lights reset and turn on one by one	The volume, mute, brightness, light thresholds, Hands-free, the Wi-Fi network, the button and the sounds are saved in flash	Move the volume slider, press Reset (EN) on the board, and check that the volume stayed	
11	Working entirely without the computer	Status sounds (offline / online / a failed question), and the button sounds: start (high) and rejected (low)	Hands-free, the BOOT button and the status sounds are saved on the board. A power bank is enough	In On the board itself, turn on BOOT button and Status sounds, and turn on Hands-free. Disconnect the USB cable from the computer, and only after that connect the board to a power bank (never both together, per the course's safety instructions). The board restarts; wait about 10 seconds for the opening greeting, and ask by voice or by pressing BOOT	
12	A light value at the extremes (complete darkness, strong light)	The value and the part of the day (night / evening / day) in the control center	The part of the day is set by the saved thresholds. A value outside the range 0-6142 is rejected and the previous value stays	The sensor is simulated with the Light sensor slider, as the course lecturer instructed. Drag it to the bottom end and ask `Is it day or night`, then to the top end and ask again. In the Serial terminal type `LIGHT 99999`: rejected	

User

#	The case	Indication to the user	Recovery	How to demo it	✓
13	Noise in the room in Hands-free mode (a knock, a clap, footsteps)	The state stays Waiting for speech. If a recording started and was discarded: a note in the conversation (`too short to be speech`)	The voice detector does not send noise to OpenAI. After several noises in a row it stops listening for a few seconds	Turn on Hands-free, knock on the table and clap next to the board	
14	Silence	Hands-free: nothing happens. Talk: the assistant answers with a generic sentence	Silence is not sent in Hands-free mode	Turn on Hands-free and stay silent for a few seconds, then turn it off, click Talk and stay silent	
15	Talk while the assistant is still answering	The state is Speaking and the button shows **Busy**. `LISTEN ignored - still waiting for or playing back a reply` , and during a greeting, a timer message or an answer to a typed question: `LISTEN ignored - the assistant is still answering, try again when it is Ready` . A rejected BOOT press plays a short low tone	The request is rejected (Talk, `LISTEN` from the terminal, and BOOT during a greeting, a timer message or an answer to a typed question) and nothing breaks. Wait for the assistant to finish speaking, or click Cancel. BOOT in the middle of an answer to a question asked with LISTEN is not rejected but cancels it, like Cancel	Ask a question, and while the assistant is answering click Talk and press the space bar	
16	The user changes their mind in the middle	Cancel (or BOOT again): `LISTEN #N CANCELLED`	The recording is deleted at OpenAI, or the answer stops, and the board goes back to Ready	Click Talk and immediately Cancel, or ask a question and click Cancel while the answer is on its way	
17	A short pause in the middle of a sentence	No special indication. One complete sentence appears in the conversation	The recording continues until about a second of silence (can be changed with Stop after silence)	Turn on Hands-free and say `What is the capital... of France` , with a half-second stop in the middle. It is sent as one question	
18	A request the board cannot carry out	The assistant says it cannot	The model knows only the tools we defined, and nothing is triggered	Say `Make the light blue` (the LED is single-color)	
19	A value out of range	The assistant says the value that was actually set	Volume set by voice is limited to 10%-100% (so the assistant can confirm out loud), and a timer to 3600 seconds. A tool that received an invalid value answers with an error message instead of staying silent	Say `Set the volume to 150 percent` and see 100%, then `Set the volume to 0` and see 10%	
20	Several requests in one command	A tag appears in the conversation for each tool that was triggered (for example LED and volume), followed by one spoken confirmation	The assistant carries them all out, and waits for all the tools to finish before the spoken answer	Say `Set the light brightness and the volume to 100 percent`	
21	A room too noisy for speaking	The typed question appears in the conversation as a bubble. A question that is too long is not sent, and the message `Too long - the board takes 200 bytes (about 200 English or 100 Hebrew letters).` appears	Ask in writing: the field under Talk (`ASK`). Up to 200 bytes (about 200 letters in English, about 100 letters in Hebrew). Anything longer is cut off, without cutting a letter in the middle, with a line in the log saying it was cut. The control center refuses to send a question that is too long	Type `What is the capital of France` in the Or type a question… field and click Ask. The assistant answers out loud	

#	The case	Indication to the user	Recovery	How to demo it	✓
22	Invalid input in a command (control center or Serial)	A message that explains what was expected, and the control goes back to the real value	The command is rejected, and nothing changes	In the Serial terminal type `VOLUME abc`, or enter a 5-character Wi-Fi password in the Wi-Fi network card	
23	The assistant hears itself	After the answer the board goes back to Waiting for speech and does not start recording	The microphone does not listen during the answer and for 0.6 seconds after it	Turn on Hands-free and ask `Tell me about the moon`. No recording starts on its own during the answer or after it	
24	Speech from a distance of a meter with background noise	The assistant carries out the request and the noise does not interfere	The voice detector tells speech from noise apart, and does not start a recording from the noise	Turn on Hands-free. One person drums on the table and the other stands a meter away and says `Turn on the light`. The LED turns on	

Stability and recovery

#	The case	Indication to the user	Recovery	How to demo it	✓
25	Asking the time before the board's clock is synchronized	At startup the OpenAI light is orange and shows `Syncing the clock…`. If there is no answer from the time servers, the OpenAI light is red, `no answer from the time servers - still trying`. And if the time is asked anyway while the clock is not valid, the assistant says that the board's clock is not set, so the time is unknown	The time is never guessed. The connection to OpenAI (and the greeting too) starts only after the clock is set, because without a clock there is no TLS. The timer runs on the board's internal counter, so it works even without synchronization	Press Reset (EN) and watch Board health: the OpenAI light is orange with `Syncing the clock…` until the clock is set, and only then does OpenAI connect	
26	A flood of commands: repeated Talk clicks, BOOT presses, dragging a slider, command lines in the terminal one after another	Talk shows `Starting…` and then `Busy`. In the terminal `LISTEN ignored - ...`, and a rejected BOOT press plays a low tone	Talk is locked until the board answers, and the board carries out one command per loop iteration and rejects LISTEN when it is busy. Sliders are sent only when the dragging stops (250ms). A full audio buffer does not overflow: the board temporarily stops reading from OpenAI or discards a chunk instead of getting stuck. Nothing breaks	Click Talk and press BOOT several times quickly, and drag the volume slider back and forth	
27	The user walks away in the middle (clicks Talk and stays silent, starts entering a network and leaves, or stops speaking in Hands-free)	The recording ends on its own: after 3 seconds with Talk, and in Hands-free after about a second of silence and in any case within 10 seconds. A network that was not confirmed: `The board didn't confirm the new network - check the serial terminal.`	Every wait has a timeout: 15 seconds without a message from the server fail the question, a recording download that stalls for 2 seconds is canceled, and a network confirmation that did not arrive is cleared after 4 seconds (up to 40 seconds while the board is still in its startup Wi-Fi wait). A form that was not sent changes nothing on the board. The board goes back to Ready	Click Talk and stay silent, and start typing a network in the Wi-Fi network card without clicking Save. The board goes back to Ready on its own and nothing is sent	

#	The case	Indication to the user	Recovery	How to demo it	✓
28	The board resets or does not respond while the control center is open	Reset: `The board restarted.`, the lights go back to Waiting and the timer chip is cleared. A board that does not report: the board prints `@@ALIVE` every 5 seconds, and after 15 seconds without a line from the board the status turns gray and `Board not responding - no report for 15 s. If it stays like this, press Reset (EN) on the board.` appears	The next line from the board restores the status. After a reset the control center rebuilds the state from the startup report, and the settings come back from flash. The alert does not run while a recording is downloading or during Replay	Press Reset (EN) on the board while the page is open. A stuck board cannot be caused on purpose: explain `Board not responding` verbally	
29	A microphone that is too loud (clipping)	No note on the page. The board counts the clipped samples, and the number appears in the timing line of each question (`clipped=` in `@@TIMING`)	The microphone's software gain was lowered from x4 to x2 after the measurement showed clipping. Clipping stays rare (1 question of 26 in the measurement)	Ask in a close, loud voice, and in the log look at `clipped=` in the `@@TIMING` line	
30	An SD card removed while the board is already running	After a recording fails to write, the board checks the card at the next free moment, and if it does not answer the SD light turns red: `card removed or failed - insert it and press Reset` . The recordings tab: `no SD card`	The assistant keeps answering, just without recordings. Insert a FAT32 card and press Reset (EN). Putting the card back without a Reset leaves the light red	Ask a question, remove the card and ask another question or two. The SD light turns red	
31	An invalid timer request (over an hour, negative, or shorter than a second)	Over an hour: the timer is set to one hour and the assistant says the limit was applied. Negative or shorter than a second: the assistant gets an error and asks again. 0 cancels a timer	A timer that is already running is not affected by an invalid request. Only 0 cancels it explicitly	Say `Set a timer for two hours` and hear that one hour was set and the assistant says this is the limit (then `Cancel the timer`). You can also try `Set a timer for half a second` : the assistant asks again	
32	A BOOT press that the board cannot carry out (offline, not ready, busy answering, saving the last answer, microphone in use)	A short low tone, always, even when Status sounds is off. A press that starts a recording plays a short high start tone. The tones follow the volume, so they are silent when the board is muted	Nothing changes. Wait for Ready (or click Cancel) and press again	In the On the board itself card, turn on BOOT button. Turn off the hotspot and press BOOT: a low tone. When it is back to Ready, a press gives a high tone and a recording	
33	A technician check without a computer	Four beeps in the order Wi-Fi, OpenAI, SD card, microphone: high = OK, low = failed (microphone OK = the microphone signal passed the check at startup)	Works without a computer and without a network, and does not change any state. Holding the button first cancels the LISTEN that the press started	When the BOOT button is turned on, hold it for 2 seconds and hear four tones. To show a failure: remove the SD card, Reset (EN) and hold it again, and the third tone is low	

#	The case	Indication to the user	Recovery	How to demo it	✓
34	A question when there is no connection (saving it for later)	`LISTEN ignored - offline (no Wi-Fi or no OpenAI). The timer and the controls still work.` and the circle shows Offline	Questions are not queued, on purpose: an answer that arrives minutes later, out of context, would confuse more than it helps, and the rejection is immediate and visible. There is nothing to save for sending later: there is no database or cloud log, and the recordings are saved on the SD card regardless of the network (and can be downloaded offline too). The timer and the controls keep working, and the settings are saved in flash	Turn off the hotspot and click Talk or press BOOT: the question is rejected immediately. The timer keeps counting	
35	The API key was changed, or OpenAI refuses it (expired, revoked, typed wrong)	The **OpenAI API key** card shows which key the board is using (`sk-...` and the last 4 characters, `saved here` or `default`). A refused key: a red OpenAI light `API key refused (HTTP 401)` , and the circle shows Offline (`OpenAI refused the API key`). A key in the wrong format is not saved: `Not saved: OPENAI_KEY ignored - ...`	Paste a key and click Save & connect: it is saved in flash, without compiling, and the board reconnects with it right away (a question in progress fails, as when switching networks). **Back to default** returns to the key from `SECRETS.h` . A saved key that is refused twice in a row is replaced on its own by the key from `SECRETS.h` (in memory only; `@@OPENAI_KEY` shows `default` , and the saved key stays in flash), and back the same way. The key does not appear in any log	Paste a wrong key in the right format (for example `sk-` and 30 letters) and click Save & connect: after two refusals the board goes back to the key from `SECRETS.h` and the OpenAI light turns green. At the end, Back to default	
36	Playing a recording from the SD card in the control center while Hands-free is on	During playback the circle shows Ready (not Waiting for speech), and under Hands-free: `Paused while a recording plays here` . Clicking Talk, or pressing BOOT, stops the player: `Paused - the board is recording a question`	The microphone does not listen until the player stops or finishes, plus 0.6 seconds (`LAPTOP_PLAYBACK`). A Hands-free sentence that was already being recorded when the playback started is canceled (`LISTEN #n cancelled`). If the page is closed in the middle of playback, the board goes back to listening on its own at the end of the recording's length	Turn on Hands-free, select an answer in the Replies tab and play it on the computer's speakers, close to the board: no recording starts. When the playback ends, the state goes back to Waiting for speech	
37	Asking by voice to switch between Hands-free and push-to-talk	The Hands-free switch moves on its own, and a note appears in the conversation: 🎙 `Hands-free on - just start talking (by voice)` . The assistant says it switched	The switch is the same as the toggle (saved in flash). It happens only on an explicit request: a question about the mode (`What is hands-free?`), a passing mention of a button, or noise does not switch the mode. A request for the mode that is already active changes nothing	With Talk, say `Switch to hands-free mode` : the switch turns on. Then, without a button, `Go back to the button` : the switch turns off. To check the caution: `What is hands-free mode?` - the switch does not move	